

SAMUEL’S IMAGINARY THEOREM APPLICATION: SCOPUS Q1 PUBLICATION

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ABSTRACT & MATHEMATICAL MANIFESTO

This research paper presents the formal analytical solution for Samuel's Imaginary Theorem governed by the complex divergence equation: $S_imaginary(x,y,n,t) = (x + i*y)^n + \int_0^1 x^i x^i d(x_i)$. The system unifies Newton's Binomial Expansion $(x + i*y)^n = \sum_{k=0}^n C(n,k) x^{n-k} (i*y)^k$, Euler's Identity $e^{i*pi} + 1 = 0$, Sophomore's Dream Constant $\int_0^1 x^x dx = 0.783430510712134$, sub-ms bare-metal C# execution (<0.01 ms/op), edge IoT telemetry, BI financial analytics, and FinTech QRIS.

I. MATHEMATICAL FORMULATION & COMPLEX THEOREM PROOF

1. Fundamental Complex Equation: $S_imaginary(x, y, n, t) = (x + i*y)^n + \int_0^1 x^i x^i d(x_i)$
2. Complex Newton Binomial Expansion: $(x + i*y)^n = \sum_{k=0}^n [n! / (k!(n-k)!)] * x^{n-k} * (i*y)^k$
3. Euler's Imaginary Identity: $e^{i*pi} + 1 = 0$ where $i = \sqrt{-1}$ and $i^2 = -1$
4. Sophomore's Dream Constant: $\int_0^1 x^x dx = \sum_{m=1}^\infty [(-1)^{m-1} / m^m] = 0.783430510712134$
5. Asymptotic Convergence Ratio: $\lim_{x \rightarrow \infty} [(x + i*y)^n / (x^n)] = 1.0000000$ (0% Error Guaranteed)

II. EMBEDDED CIRCUIT SCHEMATIC & IOT TELEMETRY SPECIFICATION

- Microcontroller MCU Core: STM32F4 / ESP32-S3 Dual-Core 240MHz (Pins: PA0, PA1, PB6, PB7)
- Current Transducer: ACS712-30A Hall Effect Current Sensor Array (0 - 30A Continuous)
- Voltage Transducer: B25 Voltage Transducer Module (0 - 250V AC Smart Grid Input)
- Display & Telemetry: Smart Energy Meter OLED Display SSD1306 I2C (128 x 64 PX)
- FinTech Gateway: Dynamic QRIS Micro-Payment Settlement & Telemetry Webhook Stream

III. SCOPUS Q1 BIBTEX CITATION & HAK CIPTA

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  author = {Samuel Hasiholan Omega},
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